Polynomial collection on the zero fiber

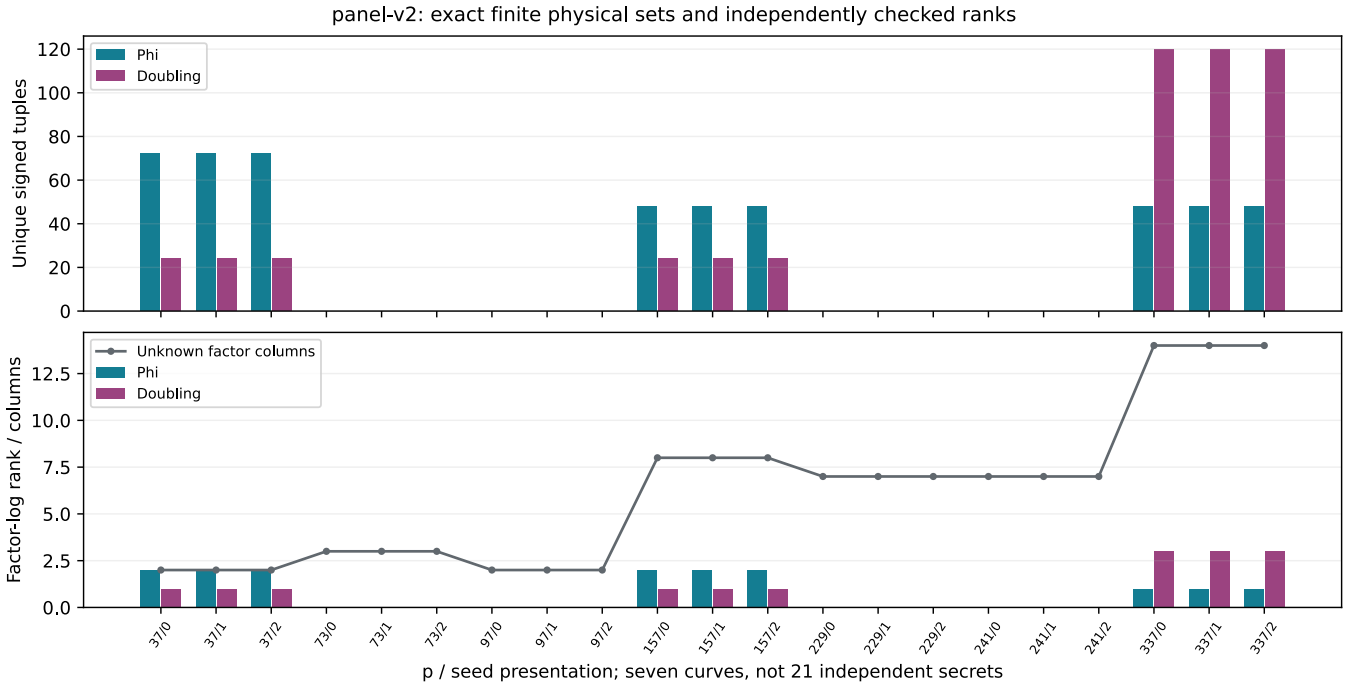
panel-v2 | 2026-10-24 | producer 57cde8bee873b43d50cf2dc4e1e74a6962604f1a

The registered 42 case-arms and their independent audits complete successfully. Each mixed arm reproduces the immutable physical census. These are case-local count sums over seven curves in three public scalar-271 presentations, not independent secret samples or an amortized DLP price.

Arm	Signed tuples	Projective rows	Zero rows	S4 systems
Phi	504	15	0	33696
Doubling	504	15	0	33696

Native case-local sums; signed tuples and projective rows are unique within each case.

The 24-action pair quotient is available to both arms. Actual lifts, chosen map, action checks, complete S4 roots/signs, point sums, folded rows and factor-point verification are executed. No factor logs, BSGS or reference sets enter collection. The separate post-run verifier reads the immutable census previously checked by point-keyed enumeration.

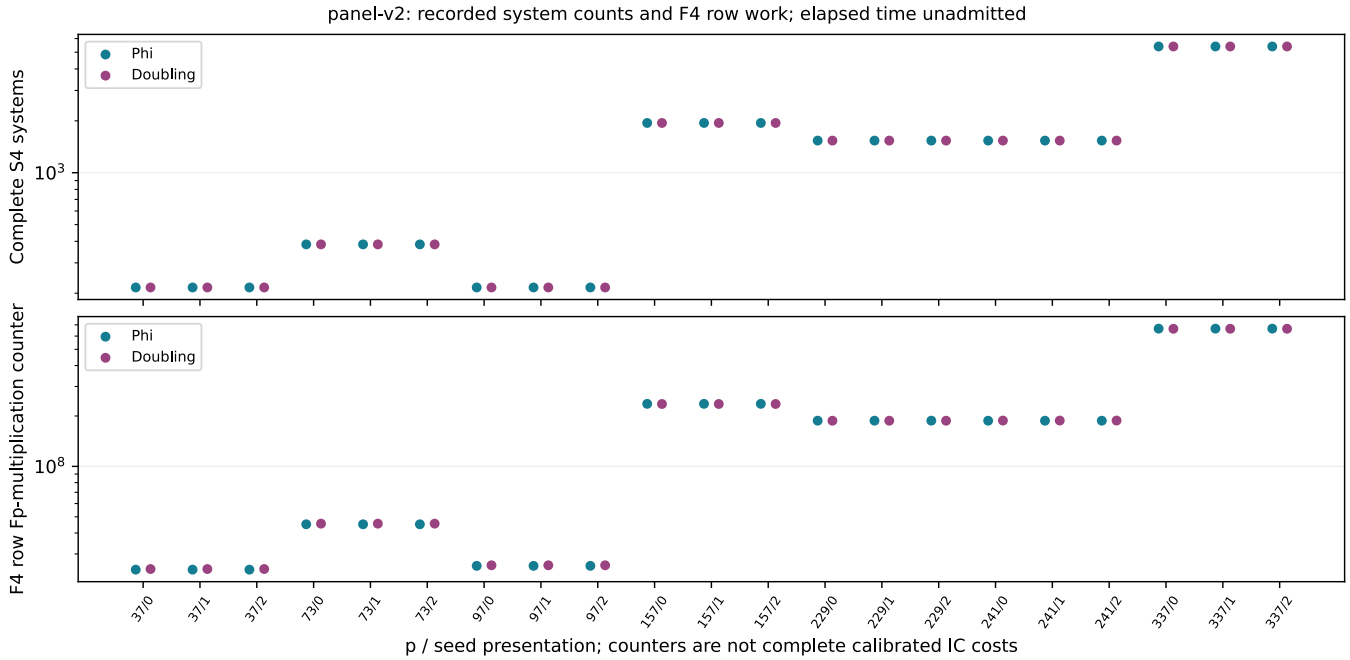


The new evidence is a complete polynomial implementation and exact physical replay of the earlier structural census. Phi determines both unanchored factor columns at p37, versus one for doubling. At p157 the ranks are 2/8 versus 1/8; at p337 they are 1/14 versus 3/14. Other tested primes give zero factor-log rank. Equal aggregate tuple/row counts do not imply equal useful rank. Transported copies supply proportional folded rows.

All registered cases

Ranks and complete system counts are copied verbatim from the native panel summary. All three presentations are shown.

p / seed	Phi tuples	Phi rank / cols	Double tuples	Double rank / cols	Phi systems	Double systems
37 / 0	72	2/2	24	1/2	216	216
37 / 1	72	2/2	24	1/2	216	216
37 / 2	72	2/2	24	1/2	216	216
73 / 0	0	0/3	0	0/3	384	384
73 / 1	0	0/3	0	0/3	384	384
73 / 2	0	0/3	0	0/3	384	384
97 / 0	0	0/2	0	0/2	216	216
97 / 1	0	0/2	0	0/2	216	216
97 / 2	0	0/2	0	0/2	216	216
157 / 0	48	2/8	24	1/8	1944	1944
157 / 1	48	2/8	24	1/8	1944	1944
157 / 2	48	2/8	24	1/8	1944	1944
229 / 0	0	0/7	0	0/7	1536	1536
229 / 1	0	0/7	0	0/7	1536	1536
229 / 2	0	0/7	0	0/7	1536	1536
241 / 0	0	0/7	0	0/7	1536	1536
241 / 1	0	0/7	0	0/7	1536	1536
241 / 2	0	0/7	0	0/7	1536	1536
337 / 0	48	1/14	120	3/14	5400	5400
337 / 1	48	1/14	120	3/14	5400	5400
337 / 2	48	1/14	120	3/14	5400	5400



Evidence boundary and next requirement



Zero-only equations have zero target coefficient. Charged target descent and complete IC pricing remain open.

Arm	Recorded F4 row Fp-multiplication counter
Phi	4112120748
Doubling	4110616998

The audit independently constructs scalar-multiplication action maps and the signed-pair partition, checks each representative's physical relations and their transported union, reconciles recorded F4 work and uses separate RREF. Public binding, incomplete systems, missing system, duplicate pair, counter mutation and nonempty-set tuple/sign mutations are rejected. Field/curve arithmetic and the exact S4 equation deriver are shared; the full curve/group and field-embedding Lean bridges remain open.

All equations are zero-only and have zero target coefficient. No target recovery or complete IC cost is established. F4 counters cover row reduction only; other field/group counters and memory telemetry are retained separately. Shared-host elapsed diagnostics include durable journaling and are not admitted timing. Calibrated 0/0.1/1 GAE, strongest matched rho, complete external-solver pricing, independent secret sampling and scaling remain open.

All failed launches and native tests remain. An outer reservation deadlocked the old panel before its first collector; zero native case records existed. That service was stopped with receipts retained. The corrected driver refuses the same misuse with exit 64. The journalled source passes 84 native tests with source checks. No completed old census, target stream or successful upload was repeated.

Remaining requirements

Requirement	Status
Complete zero-only polynomial collection	Verified on all 42 case-arms
Independent complete physical-set / matrix replay	Verified on all 42 case-arms
Charged nonzero target descent	Not yet implemented or executed
Matched complete IC / rho and calibrated price	Open
Different-twist selection and general advantage	Untested; no universal conclusion

Reproduction and provenance

Native summary SHA-256: a472fa455a2da6c691729f82b5be10f78da2cd66a700f9dba9298f763f11a2e1

Frozen production SHA-256: ba3b7c62dd51dbb158a5407e75780cd4065bbd63d9d6baa8400e5468051b4a5f

Protocol: PROTOCOL.md. Exact commands, all public inputs, independent reference hashes, per-system journals, native summary and terminal source checks are retained in the raw archive; publication details are in ARTIFACTS.md. Older 189+21 evidence remains unchanged.

The field, model, subgroup, generator, target, factor coordinates and trace are bound by each exact public-input hash and frozen source; p/seed are existing case aliases, not new curve identities.